2nd Test, Sample

1. Show how the algorithm for computing the strongly connected components of a directed graph works on the following graph.

2. How many topological orderings does the following directed acyclic graph have? Justify your answer.

3. Execute Prim's minimum spanning tree algorithm on the following graph, Let A be the starting vertex. Show the priority queue data structure at each iteration.

4. Execute Kruskal's minimum spanning tree algorithm on the following graph. Show the Union-Find data structure at each iteration.

5. Execute Dijkstra's single-source shortest path algorithm on the following graph. Let S be the starting vertex. Show the distance and parent values for each vertex at each iteration.